Project Overview Document
Executive Summary

Firefly Financial Analytics Database - Project Vision

Problem Statement

- Fragmented financial data across multiple tables
- No real-time visibility into spending patterns
- Unable to forecast cash flow or predict future expenses
- Manual reporting takes 5+ hours per week

Solution

A dedicated analytical database (star schema) connected to Firefly III that provides:

- Real-time financial KPIs
- Automated reporting
- Predictive analytics for expenses
- Self-service analytics for business decisions

Expected Outcomes

- 80% reduction in reporting time
- 30% better budget adherence through early warnings
- Cash flow predictions with 90% accuracy (after 6 months data)
- Data-driven spending decisions

Objectives & KPIs Matrix

Primary Objectives Table

Objective	KPIs	Target	Impact
Reduce operational costs	• Expense-to-revenue ratio • Cost per department • Variance vs budget	Reduce expenses by 15%	Direct profit increase

Improve cash flow visibility	• Daily cash position • Days cash on hand • Working capital ratio	Maintain 60+ days cash	Risk reduction
Predict future spending	• Forecast accuracy % • Seasonal spending patterns • Anomaly detection	90% forecast accuracy	Better planning
Optimize budget allocation	• Budget utilization % • ROI by category • Under/over performing areas	<5% budget variance	Resource efficiency

Secondary Objectives

Compliance & Governance:

- Audit trail completeness: 100%
- Transaction categorization accuracy: 95%
- Regulatory reporting readiness: Real-time

Strategic Planning:

- Growth investment identification: Quarterly
- Cost reduction opportunities: Monthly alerts
- Benchmarking against historical: Automated

Operational Efficiency:

- Report generation time: <30 seconds
- Data freshness: <1 hour
- User adoption rate: >80% of finance team

KPI Framework Document

Tier 1: Executive KPIs (Dashboard View)

```sql
-- Create a view for executive KPIs
CREATE VIEW analytical_firefly.v_executive_dashboard AS
WITH current_metrics AS (
    SELECT
        SUM(CASE WHEN transaction_type = 'withdrawal' THEN amount ELSE 0 END) AS
total_expenses,
        SUM(CASE WHEN transaction_type = 'deposit' THEN amount ELSE 0 END) AS
total_revenue,
        COUNT(DISTINCT DATE_TRUNC('month', d.full_date)) AS months_tracked
    FROM analytical_firefly.fact_transactions f
    JOIN analytical_firefly.dim_date d
        ON f.date_key = d.date_key
    WHERE d.full_date >= DATE_TRUNC('month', CURRENT_DATE) - INTERVAL '12 months'
)
SELECT
    total_revenue,
    total_expenses,
    total_revenue - total_expenses AS net_profit,
    (total_expenses / NULLIF(total_revenue, 0)) * 100 AS expense_to_revenue_ratio,
    CASE
        WHEN total_expenses > total_revenue * 0.8 THEN 'Critical - Action
Required'
        WHEN total_expenses > total_revenue * 0.6 THEN 'Warning - Review Needed'
        ELSE 'Healthy'
    END AS financial_health_status
FROM current_metrics;
```

Tier 2: Operational KPIs

KPI Category	Specific Metric	Calculation	Alert Threshold
Spending Velocity	Daily burn rate	AVG(expenses last 30 days)	>20% above 3-month avg
Budget Adherence	Category variance	(Actual - Budget) / Budget	>15% deviation
Recurring Expenses	Fixed cost ratio	Fixed costs / Total expenses	>60% reduces flexibility
Cash Runway	Months of operation	Cash balance / Monthly burn	<6 months = critical

Payment Timing	Days payable outstanding	AVG(payment date - due date)	>30 days late = risk

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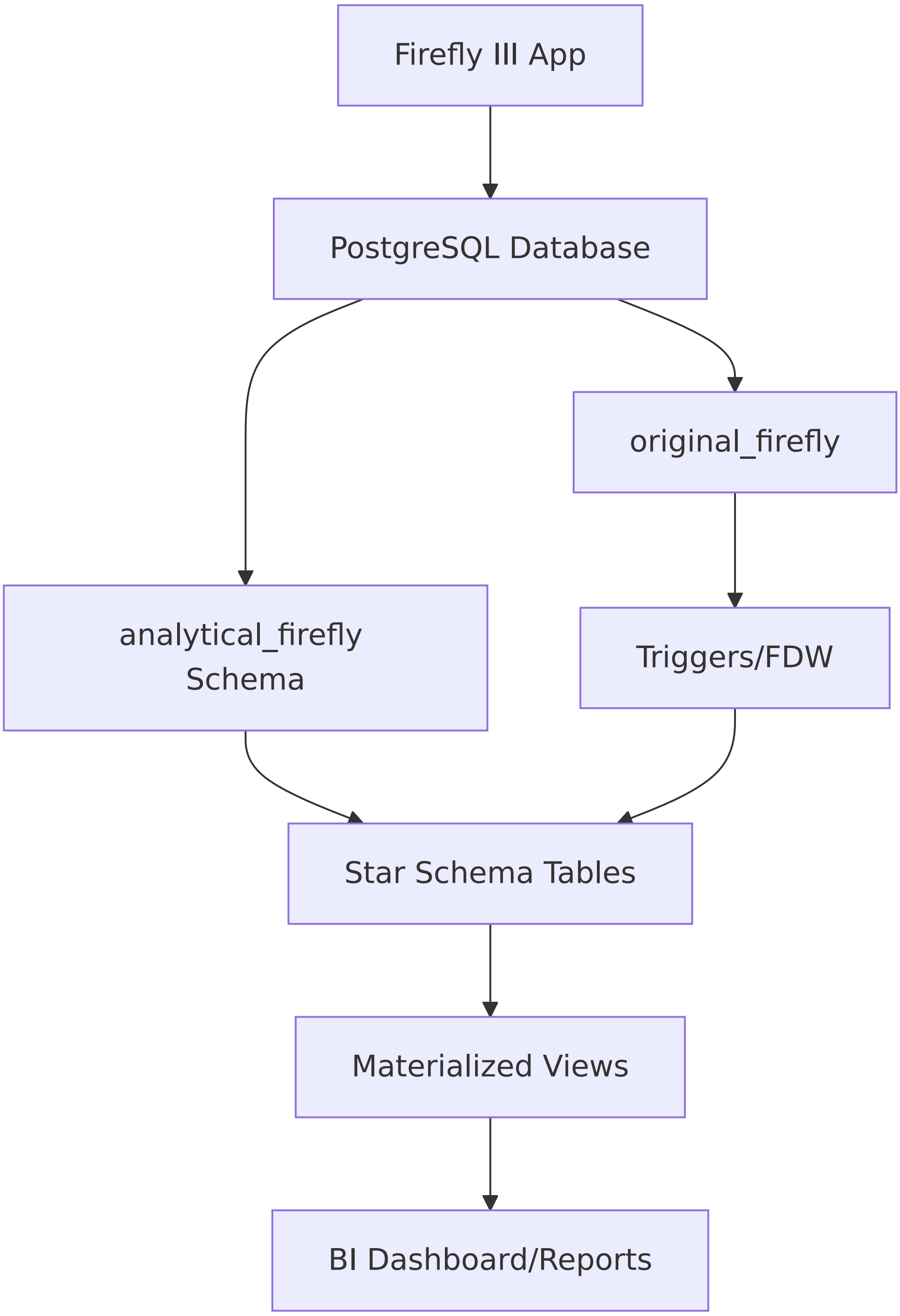
Tier 3: Predictive KPIs

```sql
-- Create a view for predictive metrics
CREATE VIEW analytical_firefly.v_predictive_metrics AS

WITH monthly_trends AS (
    SELECT
        d.year,
        d.month,
        SUM(f.amount) AS monthly_expense,

        LAG(SUM(f.amount), 1) OVER (
            ORDER BY d.year, d.month
        ) AS prev_month,

        AVG(SUM(f.amount)) OVER (
            ORDER BY d.year, d.month
            ROWS BETWEEN 3 PRECEDING AND 1 PRECEDING
        ) AS moving_avg_3m

    FROM analytical_firefly.fact_transactions f
    JOIN analytical_firefly.dim_date d
        ON f.date_key = d.date_key

    WHERE f.transaction_type = 'withdrawal'

    GROUP BY d.year, d.month
)

SELECT
    year,
    month,
    monthly_expense,

    monthly_expense - moving_avg_3m AS variance_from_trend,

    CASE
        WHEN monthly_expense > moving_avg_3m * 1.15 THEN 'Spike Detected'
        WHEN monthly_expense < moving_avg_3m * 0.85 THEN 'Dip Detected'
        ELSE 'Normal Pattern'
    END AS anomaly_flag,

    moving_avg_3m AS predicted_next_month
```

```
FROM monthly_trends;
```

Technical Architecture Document

Data Flow Diagram (Text-Based)



Component Responsibility Matrix

Component	Responsibility	Update Frequency	Owner
original_firefly	Source of truth	Real-time (user actions)	Firefly III
dim_* tables	Business entity attributes	On source change	Auto (triggers)
fact_transactions	Measurable events	Real-time	Auto (triggers)
v_* views	Pre-calculated KPIs	On query	Database
mv_* materialized views	Aggregated reports	Every 15 min	Scheduled job

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Security & Access Model

PostgreSQL Role-Based Access Control Setup

1. Create read-only role for reporting tools

```
CREATE ROLE analytics_reader;
GRANT SELECT ON ALL TABLES IN SCHEMA analytical_firefly TO analytics_reader;

-- Create power user role (finance team)
CREATE ROLE finance_analyst;
GRANT SELECT, INSERT, UPDATE ON analytical_firefly.fact_transactions TO
finance_analyst;

-- Create admin role (you)
GRANT ALL PRIVILEGES ON SCHEMA analytical_firefly TO firefly_admin;
```

Decision Support Matrix

How Each KPI Drives Decisions

Decision Type	KPI Used	Action Threshold	Business Impact
Cut expenses	Category trend	>20% MoM increase	10-15% cost reduction
Delay purchases	Cash runway	<3 months cushion	Avoids cash crisis
Increase budget	ROI by category	>15% positive variance	Optimizes resource allocation
Audit category	Anomaly detection	3 standard deviations	Prevents fraud/waste

Negotiate contracts	Vendor spend trend	Top 5 growing expenses	5-10% savings

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Predictive Alerts System

```
-- Create an alerting view for proactive decisions
CREATE VIEW analytical_firefly.v_alerts AS
WITH expense_forecast AS (
    SELECT
        d.date_key,
        SUM(f.amount) as daily_expense,
        AVG(SUM(f.amount)) OVER (ORDER BY d.date_key ROWS BETWEEN 7 PRECEDING AND
1 PRECEDING) as expected_expense
    FROM analytical_firefly.fact_transactions f
    JOIN analytical_firefly.dim_date d ON f.date_key = d.date_key
    WHERE d.full_date > CURRENT_DATE - INTERVAL '30 days'
    GROUP BY d.date_key
)
SELECT
    CURRENT_DATE as alert_date,
    'EXPENSE_ALERT' as alert_type,
    CASE
        WHEN daily_expense > expected_expense * 1.5 THEN 'CRITICAL: Expense spike
detected'
        WHEN daily_expense > expected_expense * 1.2 THEN 'WARNING: Above expected
spending'
        WHEN budget_remaining < 0 THEN 'URGENT: Budget exhausted'
    END as alert_message,
    daily_expense,
    expected_expense,
    daily_expense - expected_expense as variance
FROM expense_forecast
WHERE daily_expense > expected_expense * 1.2;
```

Implementation Roadmap

Phase 1: Foundation (Week 1)

Tasks:

- [x] Create analytical_firefly schema
- [x] Define dimension tables structure
- [x] Connect to original Firefly database

Success Metrics:

- All empty tables created
- DBeaver connection verified
- Schema documented

Phase 2: Data Population (Week 2)

Tasks:

- [ ] Fill dimension tables with historical data
- [ ] Create and populate fact_transactions
- [ ] Implement refresh mechanism (triggers/schedule)

Success Metrics:

- Data completeness: 100% of transactions
- Refresh under 5 minutes
- Zero data quality errors

Phase 3: KPI Development (Week 3)

Deliverable: 20 Core KPI Views

Example KPI View: Spending Trends

```
CREATE VIEW analytical_firefly.v_kpi_spending_trends AS
SELECT
    c.category_name,
    DATE_TRUNC('month', d.full_date) AS month,
    SUM(f.amount) AS total,
    SUM(f.amount) / NULLIF(SUM(SUM(f.amount)) OVER (PARTITION BY c.category_name),
0) * 100
        AS percentage_of_category
FROM analytical_firefly.fact_transactions f
JOIN analytical_firefly.dim_category c
    ON f.category_key = c.category_key
JOIN analytical_firefly.dim_date d
    ON f.date_key = d.date_key
GROUP BY
    c.category_name,
    DATE_TRUNC('month', d.full_date);
```

Phase 4: Dashboard Deployment (Week 4)

Dashboard	Audience	KPIs Included	Update Frequency
Executive Summary	Management	Revenue, Expenses, Profit, Cash Runway	Real-time
Operational Control	Finance Team	Budget vs Actual, Category Trends	Daily
Predictive Alerts	Decision Makers	Anomalies, Forecasts, Warnings	Hourly
Audit & Compliance	Auditors	Transaction History, Categorization	On-demand

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Phase 5: Training & Adoption (Week 5)

Training Modules

1. **Reading the Dashboards** (30 min)
    - Understanding each KPI
    - Alert thresholds and responses
2. **Running Custom Reports** (45 min)
    - Using DBeaver for ad-hoc queries
    - Exporting to Excel for analysis

3. **Action Planning** (30 min)
    - How to respond to alerts
    - Monthly review process

Success Metrics:

- 100% of finance team trained
- Weekly dashboard usage: >5x per person
- Report creation time: <10 minutes

Business Value Document

ROI Calculation Template

Estimated Annual Benefits

Direct Cost Savings

- Time saved on manual reporting: 5 hours/week $\times 50/hour \times 52 weeks =$13,000
- Early fraud detection: Estimated $5,000/year avoidance
- Optimized budget allocation: 5% improvement $\times 100,000 budget =$5,000

Indirect Benefits

- Better cash flow management: Avoid $10,000 in overdraft fees
- Negotiation power with vendors: 5-10% savings on top 3 vendors = $7,500
- Reduced audit preparation time: $3,000/year

Total Estimated Annual Benefit: $43,500

Implementation Cost

- Database setup: 20 hours $\times 75/hour =$1,500
- Dashboard development: 15 hours $\times 75/hour =$1,125
- Training: 5 hours $\times 75/hour =$375

Total Implementation Cost: $3,000

ROI = $(43,500 -$3,000) / $3,000 = 1,350% in Year 1

Sample Business Cases

Case Study 1: Preventing Cash Flow Crisis

Scenario: QuickBooks shows positive balance, but large bills due next week

Old Process:

- Manually check due dates (2 hours)
- Cross-reference bank accounts (1 hour)
- Often miss payments → late fees

New System:

1. View `v_cash_flow_forecast` dashboard
2. See color-coded alert: "Critical - Payment due in 3 days exceeds balance"
3. One-click view of upcoming obligations
4. Transfer funds or delay non-critical payments

Result: Saved $500 in late fees, avoided overdraft

Case Study 2: Identifying Wasteful Subscriptions

Scenario: Monthly expenses creeping up, no clear cause

Old Process:

- Download CSV, pivot in Excel (2 hours)
- Hard to spot recurring small charges
- Often miss unused subscriptions

New System:

1. Run `v_recurring_expense_analysis`
2. Sort by "cost per use" metric
3. Identify 5 unused subscriptions ($150/month)
4. Cancel immediately

Result: $1,800 annual savings

Maintenance & Governance

Daily Health Checks

```
-- Create a health check view
CREATE VIEW analytical_firefly.v_system_health AS
SELECT
    'Data Freshness' as check_name,
    CASE
        WHEN MAX(updated_at) > CURRENT_TIMESTAMP - INTERVAL '1 hour' THEN
'HEALTHY'
        ELSE 'STALE DATA - Check refresh job'
    END as status,
    MAX(updated_at) as last_update
FROM analytical_firefly.fact_transactions
UNION ALL
SELECT
    'Foreign Key Integrity' as check_name,
    CASE
        WHEN COUNT(*) = 0 THEN 'HEALTHY'
        ELSE 'ORPHANED RECORDS FOUND'
    END as status,
    COUNT(*) as orphan_count
FROM analytical_firefly.fact_transactions f
LEFT JOIN analytical_firefly.dim_date d ON f.date_key = d.date_key
WHERE d.date_key IS NULL;
```

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check_name	status	last_update / orphan_count
Data Freshness	HEALTHY (or STALE DATA - Check refresh job)	2026-05-07 12:34:56
Foreign Key Integrity	HEALTHY (or ORPHANED RECORDS FOUND)	0

Weekly Review Meeting Agenda

Financial Analytics Review - Weekly Agenda

Duration: 30 minutes
Attendees: Finance Team, Department Heads

1. **Dashboard Review** (10 min)
    - Review executive KPIs
    - Identify red/yellow alerts
    - Note anomalies

2. **Variance Analysis** (10 min)
    - Categories over budget
    - Unexpected spending increases
    - Forecast vs actual
3. **Action Items** (10 min)
    - Assign corrective actions
    - Update forecasts
    - Adjust budgets if needed

Quick Start Commands for You

To implement this immediately, run these commands in order:

```
-- 1. Create all the views at once
\i create_all_analytical_views.sql

-- 2. Test your first executive report
SELECT * FROM analytical_firefly.v_executive_dashboard;

-- 3. Set up automated alerts (runs every 4 hours)
SELECT cron.schedule('check-financial-alerts', '0 */4 * * *',
    'SELECT * FROM analytical_firefly.v_alerts');

-- 4. Verify everything works
SELECT
    object_name,
    object_type,
    status
FROM (
    SELECT 'v_executive_dashboard' as object_name, 'VIEW' as object_type
    UNION ALL SELECT 'v_alerts', 'VIEW'
    UNION ALL SELECT 'refresh_all', 'FUNCTION'
) checks
LEFT JOIN pg_class ON checks.object_name = relname
WHERE relname IS NULL OR relname = '';
```

Step	Description	SQL / Command

1	Create all analytical views	`\i create_all_analytical_views.sql`
2	Test executive dashboard report	`SELECT * FROM analytical_firefly.v_executive_dashboard;`
3	Schedule alerts every 4 hours	`SELECT cron.schedule('check-financial-alerts', '0 */4 * * *', 'SELECT * FROM analytical_firefly.v_alerts');`
4	Verify views and function exist	See full query in raw SQL above

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Document Deliverables Checklist

This used as project completion tracker:

Documentation Package

Vision Document (this file)

- [ ] Executive summary written
- [ ] Problem statement defined
- [ ] Expected outcomes listed

KPI Dictionary

- [ ] All 15 KPIs defined with formulas
- [ ] Alert thresholds documented
- [ ] Business owners assigned

Technical Specs

- [x] Schema design completed
- [ ] Data refresh schedule set
- [ ] Security roles defined

ROI Analysis

- [ ] Cost savings calculated
- [ ] Business cases written
- [ ] Stakeholder buy-in obtained

Maintenance Plan

- [ ] Health checks defined
- [ ] Weekly review agenda created

- [ ] Backup strategy documented